

Bank Marketing Campaign Analysis

Exploratory Data Analysis Report

A data-driven examination of customer term deposit subscriptions

1. Project Objective

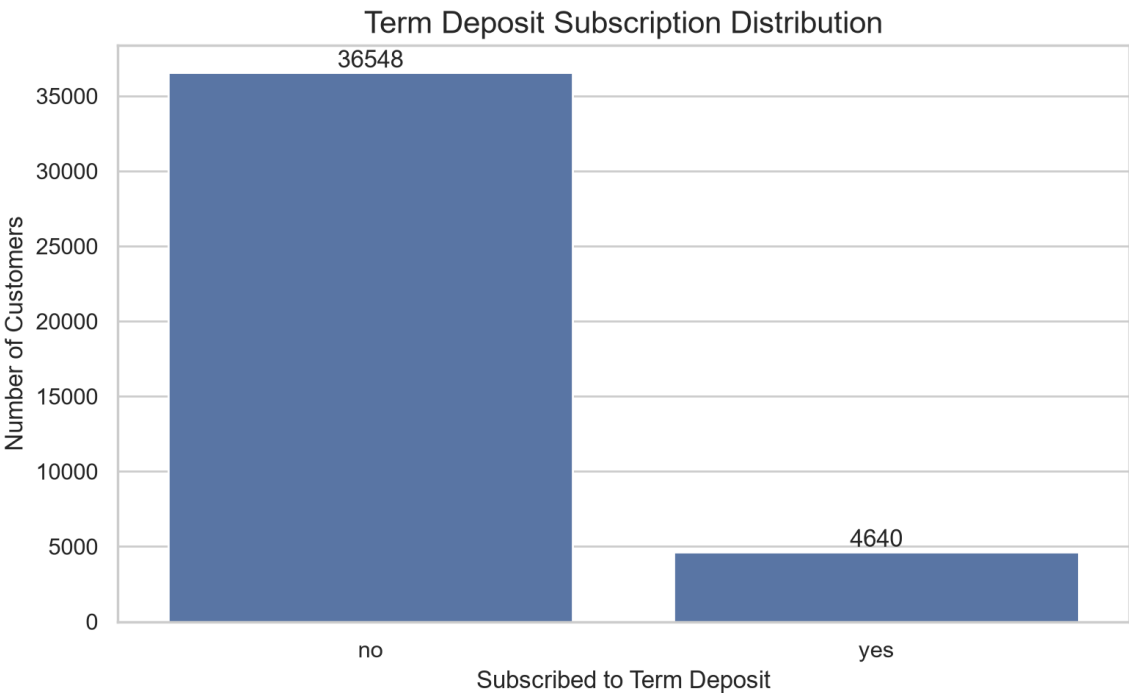
The objective is to explore customer demographics and marketing campaign characteristics associated with term deposit subscriptions. The analysis uses descriptive statistics, subscription rates and visualizations to identify patterns in the historical data.

2. Dataset Overview

Metric	Result
Total records	41,188
Total variables	21
Subscribed	4,640 (11.27%)
Not subscribed	36,548 (88.73%)
Average customer age	40.0 years

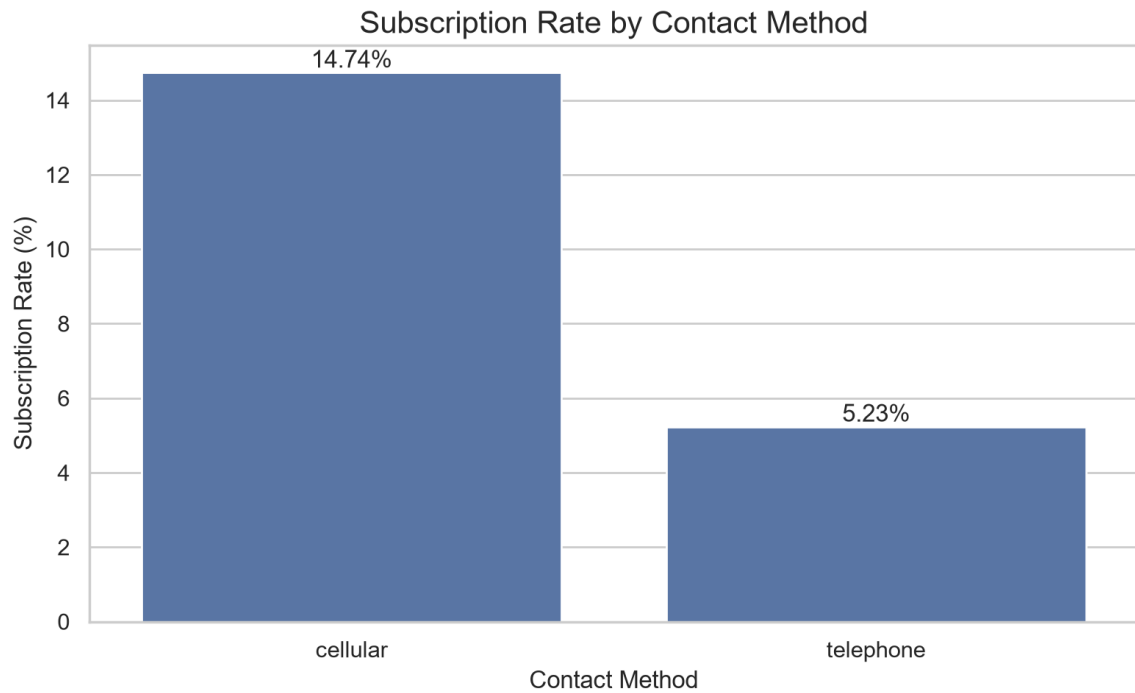
3. Subscription Distribution

Of 41,188 records, 4,640 customers subscribed, representing 11.27% of the dataset. The remaining 36,548 customers did not subscribe. This imbalance means response rates should be considered alongside raw customer counts when comparing groups.



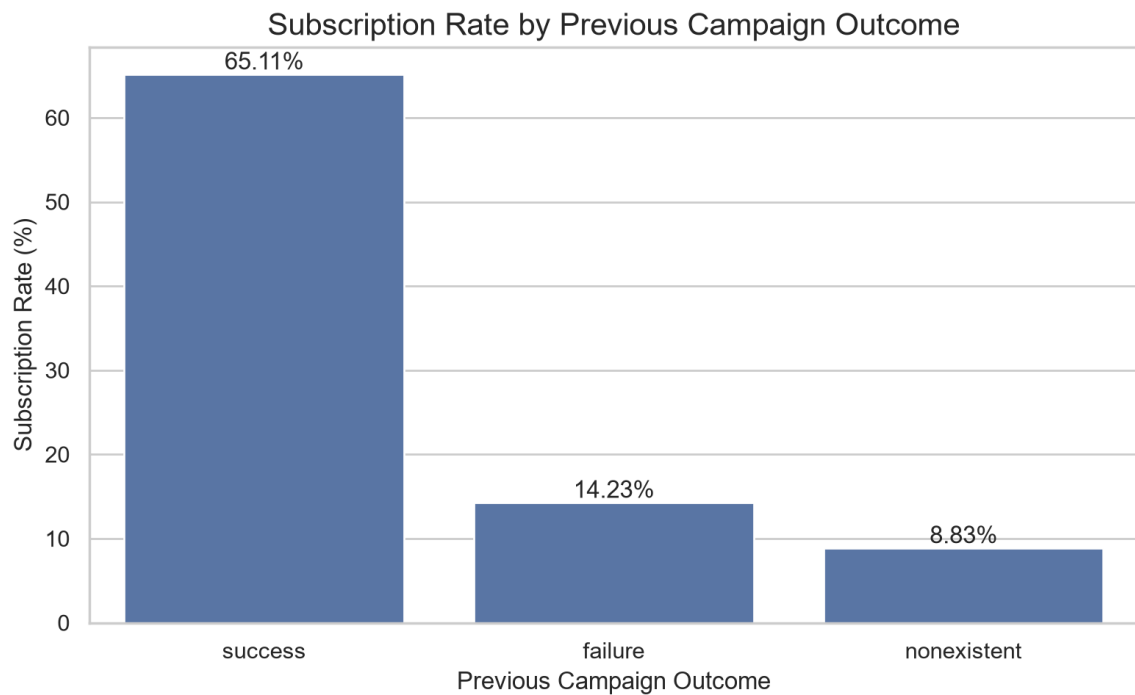
4. Contact Method

cellular: 14.74% subscription rate
telephone: 5.23% subscription rate



5. Previous Campaign Outcome

success: 65.11% subscription rate
failure: 14.23% subscription rate
nonexistent: 8.83% subscription rate



6. Additional Findings

Job categories with the highest observed subscription rates: student (31.43%); retired (25.23%); unemployed (14.20%).

Months with the highest observed subscription rates: mar (50.55%); dec (48.90%); sep (44.91%).

7. Data Quality Considerations

The dataset contains 0 actual missing values according to pandas. Some categorical values are explicitly labelled 'unknown'; these should be distinguished from true missing values when interpreting the results.

8. Limitations

These results describe associations and do not establish causation. The duration variable measures the length of the last call and is known only after the call. It should not be used as a predictor when selecting customers before contacting them.

9. Conclusion

The analysis shows that subscription rates vary across customer and campaign characteristics. These patterns can guide further investigation of campaign effectiveness. Any operational changes should be validated using appropriate experiments or newer data.

Dataset source: UCI Machine Learning Repository — Bank Marketing